

Name: Priyanshi Gupta class: SYIT Roll : S012 date of assignment:19/08/26 date pf submission:19/08/2026

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
import pandas as pd
```

```
df=pd.read_csv("Students.csv")
```

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	12	Priyanshi	BSc IT	90	95
1	6	Jueli	BSc IT	85	90
2	26	Sakshi	BSc IT	80	88
3	56	Shweta	BSc IT	88	75
4	8	Sejal	BSc IT	55	70

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
(5, 5)
```

```
print(df.dtypes)
```

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

```
print(df.shape) #number of rows and columns
```

```
(5, 5)
```

```
print(df.describe)
```

```
<bound method NDFrame.describe of
0      12  Priyanshi  BSc IT    90      95
1       6     Jueli   BSc IT    85      90
2      26    Sakshi   BSc IT    80      88
3      56    Shweta   BSc IT    88      75
4       8     Sejal   BSc IT    55      70>
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

```
result = df[df["Marks"]>=60]
print(result)
```

```
Roll No      Name Department  Marks  Attendance
0      12  Priyanshi    BSc IT    90         95
1       6     Jueli     BSc IT    85         90
2      26    Sakshi     BSc IT    80         88
3      56    Shweta     BSc IT    88         75
```

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

```
result = df.sort_values("Marks")
print(result)
```

```
Roll No      Name Department  Marks  Attendance
4       8     Sejal     BSc IT    55         70
2      26    Sakshi     BSc IT    80         88
1       6     Jueli     BSc IT    85         90
3      56    Shweta     BSc IT    88         75
0      12  Priyanshi     BSc IT    90         95
```

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks ",df["Marks"].max())
```

```
print("Minimum Marks ",df["Marks"].min())  
print("Average Marks ",df["Marks"].mean())  
print("total Marks ",df["Marks"].sum())
```

```
Maximum Marks  90  
Minimum Marks  55  
Average Marks  79.6  
total Marks  398
```